Interactive and Supervised Dual-Mode Attention Network for Remote Sensing Image Change Detection

Hongjin Ren, Min Xia, *Member, IEEE*, Liguo Weng, Haifeng Lin, Junqing Huang, *Student Member, IEEE*, and Kai Hu

***Abstract*— With the rapid advancement of remote sensing technology, change detection using bitemporal remote sensing images has significant applications in land use planning and environmental monitoring. The emergence of convolutional neural networks (CNNs) has accelerated the development of deep learning-based change detection. However, existing deep learning algorithms exhibit limitations in understanding bitemporal feature relationships and accurately identifying change region boundaries. Moreover, they inadequately explore feature interactions between bitemporal images before extracting differential features. To address these issues, this article proposes a novel interactive and supervised dual-mode attention network (ISDANet). In the feature encoding stage, we employ the lightweight MobileNetV2 as the backbone to extract bitemporal features. Additionally, we design the neighbor feature aggregation module (NFAM) to aggregate semantic features from adjacent scales within the dual-branch backbone, enhancing the representation of temporal features. We further introduce the interactive attention enhancement module (IAEM), which effectively integrates self-attention and cross-attention mechanisms. This establishes deep interactions between bitemporal features, suppresses irrelevant noise, and ensures precise focus on true change regions. In the feature decoding stage, the supervised attention module (SAM) reweights differential features and leverages supervisory signals to guide the learning of attention mechanisms, significantly improving boundary detection accuracy. SAM dynamically aggregates multilevel features, balancing high-level semantics and low-level details to capture subtle changes in complex scenes. The proposed model achieves F1 scores that are 0.28%, 1.6%, and 0.76% higher than the best comparative method, spatiotemporal enhancement and interlevel fusion network (SEIFNet), on three CD datasets [LEVIR-CD, Guangzhou dataset (GZ-CD), and Sun Yat-sen University dataset (SYSU-CD)], respectively, while maintaining a lightweight design with only 6.93 M parameters and 3.46G floating-point operations (FLOPs). The code is available at https://github.com/RenHongjin6/ISDANet.**

***Index Terms*— Bitemporal feature interaction, change detection, deep supervision, multiscale fusion.**

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I. Introduction

CHANGE detection in remote sensing images involves the identification of surface or object changes over time by comparing multitemporal remote sensing images obtained through remote sensing technology. This technique garners significant attention in the field of Earth observation and plays a crucial role in urban development planning [1], monitoring vegetation cover changes [2], [3], disaster assessment [4], [5], among other domains. However, current change detection tasks encounter several challenges, including disturbances during the acquisition of bitemporal remote sensing images due to seasonal variations, changes in lighting conditions, cloud cover, among other interfering factors, which subsequently affect the accuracy of change detection. Consequently, accurately identifying actual changes in remote sensing images affected by these interfering factors has become an exceptionally challenging task.

The methods proposed for remote sensing image change detection can generally be categorized into two types: traditional methods and deep learning-based methods. Among traditional methods, the most common approach is direct comparison, including techniques like differencing [6], [7] and thresholding [8]. These methods are simple and directly compare each pixel of bitemporal images to identify changes based on pixel differences. However, they have significant limitations, especially when dealing with complex land cover changes. Traditional methods are sensitive to image quality and noise, and can be disrupted by factors such as clouds and shadows, leading to false positives or negatives. Additionally, they typically rely on pixel-level comparison, which fails to capture global changes in the image, and may miss subtle or localized changes in large-scale or complex scenarios. These methods also require manual feature design and are dependent on complex image registration and quality correction steps, making the process cumbersome and error-prone. As a result, traditional methods often perform less efficiently and accurately than deep learning-based approaches, especially in high-precision and automated change detection tasks. In contrast, traditional machine learning methods, such as support vector machines [9], K-nearest neighbors [10], and random

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forest regression [11], attempt to overcome the limitations of direct comparison methods by using various image features, such as differential and spatiotemporal features, to distinguish changing pixels. However, these methods also face challenges. They rely on manual feature selection and design, and require high-quality image registration, which increases computational complexity and processing time when dealing with large-scale or multitemporal images. Furthermore, traditional machine learning methods often fail to capture the nonlinear and high-dimensional relationships of complex land cover changes (e.g., vegetation, water bodies, urban areas), making them less effective than deep learning approaches. Additionally, these methods struggle with capturing temporal changes and long-range dependencies, failing to model global context information, which further impacts the accuracy and efficiency of change detection.

Recent research in remote sensing image change detection has increasingly focused on deep learning methods, which offer strong nonlinear modeling capabilities [12], [13], [14], [15], [16], [17], making them well-suited for diverse change detection tasks. Moreover, their end-to-end training processes allow more efficient optimization of the entire model, making the training process more automated and effective. Consequently, deep learning-based methods have significantly improved performance in remote sensing image change detection, becoming the current research hotspot and mainstream approach. From early models such as fully convolutional network (FCN) [18], U-Net [19], DenseNet [20], PSPNet [21], and HRNet [22] to the recent emergence of Transformer networks [23], these deep learning models have demonstrated notable efficacy in remote sensing image change detection tasks. For instance, Liu et al. [24] proposed a change detection method based on multiscale feature fusion, which uses convolutional neural networks (CNNs) to extract multiscale features and fuse features from different scales for more accurate change detection. While this method achieves some improvement in accuracy, it has high computational complexity due to its reliance on multilayer CNN feature extraction, particularly when processing large-scale remote sensing images, resulting in slow training and inference speeds and being constrained by convolution kernel selection and scale settings. Fu et al. [25] introduced a dual-attention network (DANet) for scene segmentation, which models global semantic information using a self-attention mechanism. Although this method effectively captures global context, the self-attention mechanism incurs significant computational overhead, presenting challenges in resource consumption and time efficiency. Specifically, in large-scale image change detection, the computational cost of DANet may increase substantially, limiting its feasibility for real-time or large-scale applications. Furthermore, Xue et al. [26] proposed a triple-change detection network (TCDNet) that considers long-range dependencies of change objects at multiple scales to achieve more accurate change detection. While TCDNet enhances detection accuracy by introducing multilevel long-range dependency modeling, it requires longer training times and imposes high computational complexity and memory demands when processing multiple temporal images, affecting its efficiency. Additionally, multilevel Swin-Transformer networks [27] and U-shaped Swin-Transformer [28] directly perform change detection tasks using multihead attention mechanisms, excelling in capturing long-range dependencies and global context. However, the application of Swin-Transformer models in image change detection also faces significant challenges, particularly in terms of computational overhead. Swin-Transformer requires considerable memory and computational resources, and training can become slow and difficult to scale, especially when handling high-resolution images, while CNN methods have strong feature extraction capabilities and are sensitive to local features. However, CNN methods are limited in handling long-range contextual information and may, therefore, overlook global context in certain situations. In such cases, introducing attention mechanisms, such as self-attention [29], [30], cross-attention [31], [32], axial cross-self-attention [33], spatial attention [34], and channel attention (CA) [35], is a feasible approach to enhance the modeling of global context. For example, Chen et al. [36] innovatively proposed the bitemporal image transformer (BIT), which integrates CNN-extracted features to capture global information in remote sensing images, achieving satisfactory performance in change detection tasks. However, BIT's feature fusion relies on simple channel stacking, limiting deep interactions between bitemporal features. This approach may fail to capture detailed changes between different time points in complex scenarios, potentially overlooking crucial contextual information and affecting detection accuracy. Additionally, while straightforward, channel stacking falls short in modeling complex relationships and long-range dependencies, especially when handling high-resolution or multitemporal images, resulting in limited performance. Zhang et al. [37] attempted to align bitemporal features at the channel level to reduce irrelevant factors, achieving some improvements. However, this method is relatively simplistic, primarily relying on channel-level alignment, and does not effectively capture more complex, nonlinear relationships between features. In intricate change scenarios, channel alignment alone may miss deeper connections, impacting detection accuracy and robustness. Feng et al. [38] facilitated feature alignment and fusion by exchanging query vectors Q between bitemporal features, promoting interaction to some extent. Nevertheless, solely relying on query vector Q exchange for feature interaction is insufficient, as it fails to fully explore the complex relationships between features. While this approach improves model performance to some degree, it lacks a more comprehensive interaction mechanism, falling short of the efficiency required for effective change detection. Enhancing self-attention and cross-attention mechanisms by incorporating richer interaction information is necessary to improve model effectiveness and efficiency.

Overall, the aforementioned approaches to change detection exhibit certain limitations. First, prior deep learning-based change detection methods have overly focused on extracting features from bitemporal remote sensing images, thereby neglecting the study of spatiotemporal changes [39]. Seasonal variations, causing significant alterations in surface features like vegetation and water bodies, have disrupted the accuracy of change detection. Additionally, over time, human activities

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and natural factors lead to the evolution of surface features, making the precise identification of genuine change areas more complex and challenging [40]. Hence, a model centered around capturing spatiotemporal variations is imperative. Primarily, the distribution of change areas in bitemporal remote sensing images invariably necessitates the incorporation of bitemporal features to capture them. Consequently, addressing the impact of spatiotemporal transformations on remote sensing image change detection requires robust and efficient feature interactions within bitemporal remote sensing images and the capture of global differential features. This facilitates the comprehensive propagation of spatial information within bitemporal features, enabling a better understanding of whether the objectives related to change detection are significant. In particular, during the encoding stage, the backbone network extracts rich bitemporal semantic features, but features at different scales have different semantic information. Therefore, we consider interacting and fusing features of adjacent scales in the backbone network before performing bitemporal feature interaction during the encoding stage. This can provide richer and more comprehensive semantic information for the subsequent bitemporal feature interaction. Additionally, designing a good decoder can better utilize the features extracted during the encoding stage and can better identify the edge details of change areas and small object features. However, existing decoder structures have limited capability in identifying boundaries of change areas and capturing small objects during the progressive upsampling process due to the attention strategies used. Yin et al. [41] used a weighted fusion of spatial attention and CA to hierarchically refine the extracted features and reconstruct the change map. However, this fusion did not enhance the focus on the edge details of change objects. Lv et al. [42] proposed a hierarchical attention fusion strategy that effectively captures global semantic features from multiple perspectives, addressing the shortcomings of existing networks in feature attention and fusion. However, its main limitation lies in the high computational complexity, especially when handling high-resolution or large-scale data, resulting in significant training costs.

This article proposes the interactive and supervised dual-mode attention network (ISDANet) to address the limitations of existing methods in bitemporal remote sensing image change detection, particularly in feature fusion and interaction. The innovation lies in effectively combining self-attention and cross-attention in the encoding phase to enhance the interaction of multiscale features, improving the representation of change regions in bitemporal remote sensing images. In the decoding phase, we introduce a supervised attention mechanism to progressively weight and fuse features, effectively improving boundary detection and small target accuracy. First, ISDANet uses the pretrained MobileNetV2 [43] to extract bitemporal features, which has demonstrated excellent performance in tasks such as image segmentation [44] and object detection [45]. Compared to traditional heavy networks like ResNet-34 [46] and VGG19 [47], MobileNetV2 offers significant advantages in computational efficiency and feature extraction, especially in resource-constrained environments. By selecting MobileNetV2, we ensure good feature extraction performance while maintaining computational efficiency. Next, to effectively integrate multiscale and multilevel features, ISDANet introduces the neighbor feature aggregation module (NFAM), which aggregates semantic features from adjacent scales, further enhancing the representation of temporal features. Unlike traditional feature fusion methods, NFAM optimizes the aggregation of multiscale features, improving model accuracy and robustness. For interactive information, ISDANet designs the interactive attention enhancement module (IAEM), which effectively combines self-attention and cross-attention. IAEM enables feature sharing and interaction across different time points, suppressing noise interference, and sensitively capturing global semantic features of change regions. Finally, to improve boundary detection and small target accuracy, ISDANet introduces the supervised attention module (SAM) in the decoding phase. SAM recalibrates and fuses multilevel features, achieving progressive feature learning from coarse to fine, ensuring fine-grained feature representation while maintaining global context consistency. The main contributions of this article can be summarized as follows.

1) The proposed ISDANet framework unifies self-attention and cross-attention to merge multiscale features from bitemporal remote sensing images. During decoding, it guides supervised attention from coarse to fine, progressively aggregating features. This approach enhances target region and edge detail restoration while reducing missed detections and false alarms. The network is end-to-end trainable, simplifying the process.
2) The proposed NFAM aggregates semantic features from adjacent scales, enhancing the backbone's bitemporal feature representation. The IAEM combines self-attention and cross-attention, enabling feature interaction across time to highlight real changes and suppress noise. The SAM progressively fuses multiscale features during decoding, enhancing fine-grained representation while maintaining global context consistency.

   We tested ISDANet on three datasets, and it outperforms other state-of-the-art (SOTA) methods in terms of accuracy and robustness, while maintaining lightweight computational costs.

The rest of this article is organized as follows. Section II provides a brief review of related work. Section III describes the details of the proposed method. Section IV presents a series of experimental results and detailed analyses. In Section V, we summarize the work presented in this article.

II. Related Work

A. CNN-Based Methods

Recently, CNN-based change detection methods have embraced novel strategies to address the historical issue of suboptimal feature extraction across different scales. Among these, the multiscale feature fusion strategy [48], [49], [50] stands out as it amalgamate multiscale contextual semantic information, facilitating the localization of target features in changing areas. For instance, Xu et al. [49] utilized a feature

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pyramid to enable feature extraction and reinforced fusion at diverse hierarchical levels, enhancing the comprehensive understanding of semantic features within images. Ma et al. [51] employed multiscale stripe convolution to extract various scale features from images, achieving fusion among multiple features and obtaining a finer-grained representation of features. Furthermore, to further augment the feature representation in the change domain, some attention-based mechanisms [49], [50], [52] have been proposed. The introduction of attention mechanisms enables adaptive capturing of change regions by the model and suppresses irrelevant noise.

It is worth noting that CNNs exhibit strong capabilities in local feature extraction, effectively capturing multilevel features within images, including image edges, textures, and other low-level features, as well as higher level features. However, CNN methods lack the ability to model long-range contextual semantics, potentially leading to the oversight of global contextual semantic information in certain scenarios. Hence, this leads to the introduction of the transformer-based CD methods that will be discussed below.

B. Transformer-Based Methods

The above has mentioned that CNN methods are limited in handling long-range contextual information, whereas Transformers can comprehensively capture global relationships among spatial and temporal features in remote sensing images through self-attention mechanisms, enabling the model to effectively comprehend change-related information in the images. Moreover, Transformers can utilize positional encoding to provide location information for each position, preserving the relative spatial relationships between pixels in remote sensing images, aiding the model in better understanding the spatial structure of the image. Particularly, Transformer-based models have shown robust performance on large-scale datasets, such as in tasks like object detection [53] and semantic segmentation [54]. Given the unique capabilities of Transformer models, they hold great potential for change detection tasks as well.

In contrast to CNN models, which focus on modeling local detailed information, Transformer models excel in capturing global relationships. Chen et al. [36] introduced BIT, a lightweight architecture that models spatiotemporal contexts but relies on simple channel stacking, limiting feature interaction between dual-temporal remote sensing images. Song et al. [33] proposed a dual-branch structure combining CNN and Transformer in parallel: CNN extracts fine-grained features, while Transformer captures global information. These are fused through a bidirectional interaction structure, though at the cost of higher memory and computational requirements. Similarly, Chu et al. [55] utilized a dual-branch structure with CNN and Transformer in each branch to guide mutual feature exploration for bitemporal images, reducing false positives and negatives but increasing memory consumption and computational overhead. Yuan et al. [56] proposed a novel semantic change detection (SCD) model that selectively merges semantic tokens in the multihead self-attention block to obtain multiscale features, which is vital for detecting changes at different scales in remote sensing images.

C. Attention-Based Methods

The use of attention mechanisms aims to dynamically assign different weights to different parts of the input data, allowing the model to focus its attention on the most crucial information for the current task. In the context of change detection tasks, attention mechanisms can automatically learn and allocate weights to changed and unchanged regions, enabling the network to assign greater importance to pixels in changed regions while suppressing interference noise from unchanged regions, thereby better capturing change information. Nowadays, many attention allocation strategies have emerged. Liu et al. [57] proposed position attention modules and CA modules, which can adaptively select and emphasize building features with high semantic responses. However, this attention allocation strategy is overly simplistic and may lead to loss of features related to small targets. Song et al. [58] introduced a high/low frequency (HiLo) attention mechanism to extract more detailed local features, and used a multiscale pooling pyramid transformer (MPPT) module to focus on more global features, but this approach requires significant computational resources. Lu et al. [59] utilized a multiscale CA mechanism combined with a multiscale feature fusion strategy for attention allocation to improve performance, but relying solely on CA may overlook edge details of changed regions. Song et al. [50] introduced spatial attention mechanisms in both encoding and decoding stages, allowing the model to focus on spatial attention allocation throughout the process, leading to a performance improvement, but this type of attention allocation is too singular and cannot accurately identify edge details and features of small targets in changed regions.

III. Methodology

In this section, we first provide an overview of the proposed method. Subsequently, we delve into the specifics of NFAM, IAEM, and SAM.

A. Network Architecture

The ISDANet proposed by us is a weight-shared, dual-branch, end-to-end trainable network, based on an encoder–decoder structure, as depicted in Fig. 1.

1) *Encoder:* Previous studies have demonstrated the effectiveness of CNN methods in image feature extraction [60], [61], [62]. Compared to some commonly used but more complex backbones like ResNet-34 [46] and VGG19 [47], we employ the lightweight backbone network MobileNetV2 [43] to extract bitemporal features. We remove the original global average pooling layer and the last fully connected layer from the backbone to adapt it to our remote sensing image change detection task. Specifically, we extract features from the bitemporal remote sensing images using two parallel branches of the pretrained backbone network (MobileNetV2). As each stage of the backbone contains a convolutional layer with a stride of 2, the feature map sizes extracted from the five stages of the dual-branch backbone are 128×128, 64×64, 32×32, 16×16, and 8×8, respectively.

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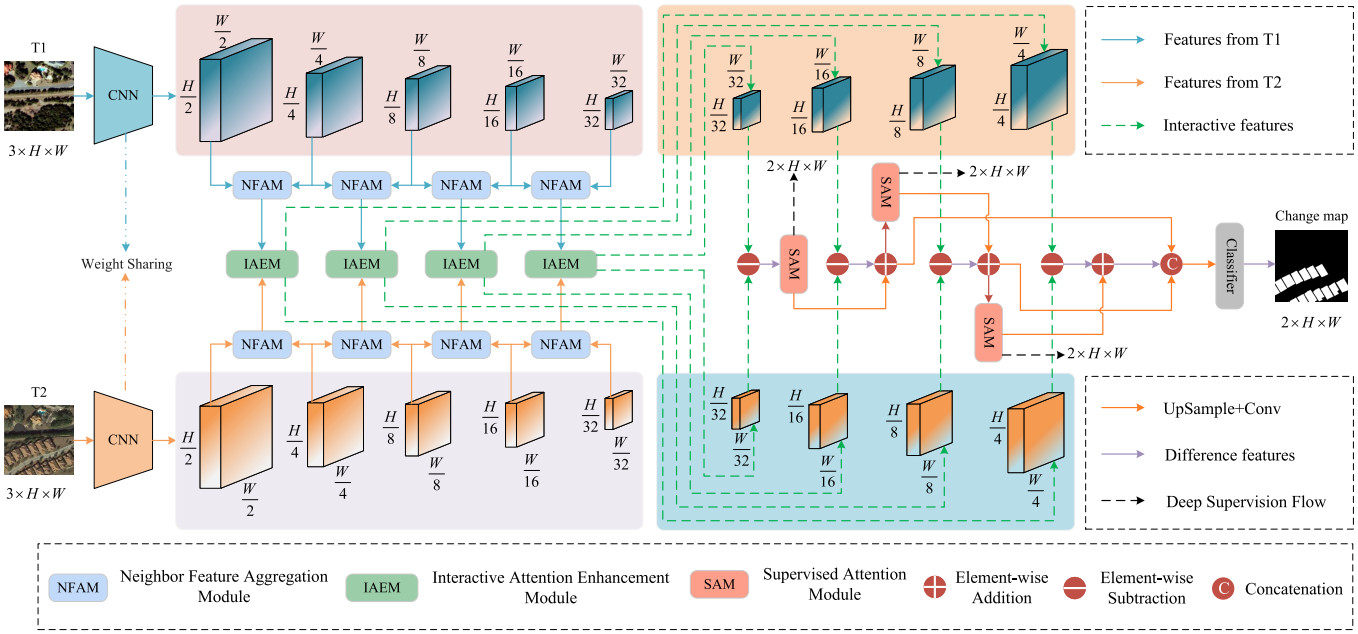


Fig. 1. Proposed ISDANet framework. In the encoding stage, the Siamese backbone network MobileNetV2 progressively extracts features from input bitemporal remote sensing images, while NFAM enhances the representation capability of the extracted features. IAEM facilitates comprehensive interaction and coupling of bitemporal features from different temporal distributions, followed by pixelwise subtraction to obtain the change differences. In the decoding stage, SAM performs reweighted fusion of differential features, enabling progressive feature aggregation from coarse to fine. Finally, different-scale features from the decoding phase are concatenated and fused to generate the ultimate change map.

To fully leverage the semantic features extracted at different scales and levels from the backbone, we propose an NFAM to enhance the representation capability of bitemporal features. This module facilitates the sharing and fusion of semantic features from different scales and levels of the dual-branch backbone, enhancing object feature representation and granularity. Furthermore, to enable comprehensive interaction and coupling of the bitemporal features, we introduce IAEM. Compared to JointAtt proposed by Feng et al. [38], our approach incorporates more interactive information to effectively unify self-attention and cross-attention mechanisms. This enables more comprehensive sharing and interaction of features distributed across different time periods, suppresses irrelevant noise interference, and more sensitively captures global semantic features across different time points. More importantly, by adopting a more strongly coupled interactive attention mechanism, we have obtained a richer attention distribution for the interaction of dual-temporal features, significantly enhancing the perceptiveness and accuracy of ISDANet toward complex scene changes, particularly concerning edge details of genuine change objects and some easily overlooked small target features.

2) *Decoder:* In the decoding stage, we designed an SAM to explore spatial and channel information and provide additional supervisory signals to the network. Utilizing the multilevel differential features obtained from the encoder, we incorporate SAM for staged upsampling and multilevel feature fusion, allowing for a more accurate and detailed prediction of change maps. Additionally, we further fuse the multiscale features, which have undergone weighted fusion from coarse to fine through SAM, thereby enhancing fine-grained representation while preserving consistency in the global context.

B. Neighbor Feature Aggregation Module

As mentioned earlier, MobileNetV2, being a lightweight network, exhibits commendable feature extraction capabilities. However, the features extracted at different stages in MobileNetV2 are mutually independent, encompassing both high-level semantic information and low-level semantic information. High-level semantic features generally refer to abstract features associated with objects, scenes, or overall structures. In change detection tasks, these might include geographical elements such as buildings, roads, and vegetation. Conversely, low-level semantic features are related to localized information like details, textures, and other local attributes. Considering that the multiscale, multilevel features extracted by the backbone network encapsulate diverse semantic information about various aspects of objects, we propose NFAM to integrate these multiscale, multilevel semantic features. This integration aims to further strengthen the detailed information in low-level features and augment the contextual information in high-level features. Moreover, it facilitates the sharing and fusion of these two types of features, enabling the model to comprehend image information more comprehensively.

The proposed NFAM is illustrated in Fig. 2. This module takes inputs from the first to the fifth stages of the backbone, with adjacent stages being fused for each of the five stages. Taking the T1 branch as an example, let us define the input feature Input1 as $X_1 \in \mathbb{R}^{C\times H\times W}$ and Input2 as $X_2 \in \mathbb{R}^{C\times H\times W}$.

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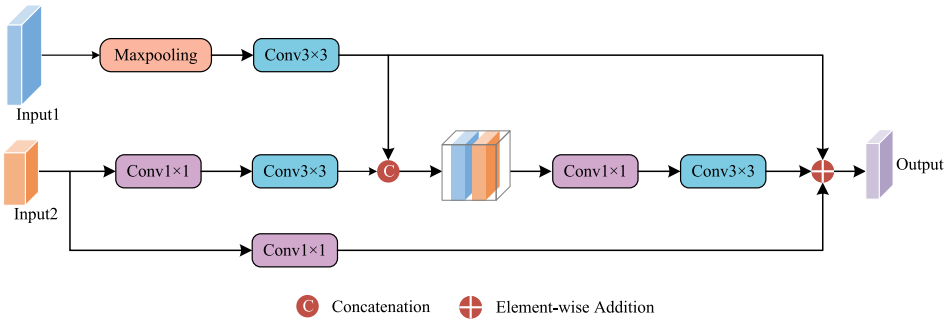


Fig. 2. Illustration of the proposed NFAM.

First, X_1 undergoes a max pooling operation to downsample its resolution to match that of X_2. Subsequently, a 3×3 convolution is applied with batch normalization and ReLU activation functions to increase the channel dimensions. We employ a 1×1 convolution to adjust the channels of X_2 and utilize a 3×3 convolution for further feature extraction. The process can be formulated as follows:

$$X_1{}' = \text{Conv}_{3\times3}(\text{Maxpool}(X_1)) \tag{1}$$

$$X_2{}' = \text{Conv}_{3\times3}(\text{Conv}_{1\times1}(X_2)) \tag{2}$$

where MaxPool$(\cdot)$ denotes the max pooling operation, $\text{Conv}_{1\times1}(\cdot)$ represents a 1×1 convolutional layer with a batch normalization and a ReLU activation function, and $\text{Conv}_{3\times3}(\cdot)$ denotes a 3×3 convolutional layer with a batch normalization and a ReLU activation function.

After the aforementioned feature transformations, the fusion of $X_1{}'$ and $X_2{}'$ is achieved through concatenation, aggregating multiscale information. We adjust the channels of X_2 through a 1×1 convolution and retain its original features through a residual connection. The channel of the concatenated features is adjusted using a 1×1 convolution, followed by feature extraction using a 3×3 convolution. The process can be formulated as follows:

$$Y = \text{Conv}_{1\times1}(X_2) + \text{Conv}_{3\times3}(\text{Conv}_{1\times1}(\text{Concat}(X_1{}', X_2{}'))) + X_1{}' \tag{3}$$

where Concat$(\cdot)$ represents the feature splicing in the channel dimension.

C. Interactive Attention Enhancement Module

Bitemporal remote sensing images are captured images of the same geographical area at different time points. However, semantic differences between bitemporal features are often influenced by various natural factors such as seasonal changes, alterations in lighting conditions, and modifications in terrain and topography. Therefore, research into the interaction of bitemporal features becomes crucial. Since our focus is primarily on genuinely changed areas within bitemporal remote sensing images, which typically constitute only a small portion, enabling mutual guided interaction between these images to facilitate the model in generating attention distributions focused solely on authentic changed regions is a feasible approach. Zhang et al. [37] aligned bitemporal features at the channel level to reduce the interference of irrelevant noise. However, this alignment mainly focuses on the channel dimension, limiting the feature representation capability and failing to capture more complex relationships between features. Feng et al. [38] attempted to directly align and fuse features by swapping the query vectors Q, achieving some degree of interaction between bitemporal features. However, relying solely on swapping the query vectors Q for feature interaction means that the interaction occurs only in a single direction, with shallow interaction depth. This approach does not fully consider the similarity between keys K and queries Q, failing to fully utilize the complex relationships between features. We propose a new interaction module, the IAEM, which performs feature interaction in both directions through bidirectional interaction between queries Q and keys K. This bidirectional interaction captures more comprehensive feature information. Moreover, the complementary fusion of bidirectional interaction features allows for better integration of bitemporal features, enhancing feature representation capability and change perception. The proposed IAEM is shown in Fig. 3.

The input features are the enhanced features $X_1, X_2 \in \mathbb{R}^{C\times H\times W}$ from bitemporal images processed by NFAM. Initially, the input features undergo a 1×1 convolution followed by a depthwise separable convolution (DSConv) [63] separately to generate query (Q), key (K), and value (V). Specifically, $Q_1{}'$ and $Q_2{}'$ are concatenated to form Q_{cat}, which undergoes elementwise multiplication and Softmax operation with its corresponding $K(K_1, K_2)$, followed by multiplication with the respective $V(V_1, V_2)$. Since Q_{cat} simultaneously encompasses complementary queries for bitemporal features, it enables the model to comprehensively consider input features from different time points. Similarly, concatenating $K_1{}'$ and $K_2{}'$ produces K_{cat}, which undergoes elementwise multiplication and Softmax operation with their respective $Q(Q_1, Q_2)$, followed by multiplication with their respective $V(V_1, V_2)$. Hence, K_{cat} contains key information from the bitemporal features. Finally, we reshape the two different interaction features obtained from both branches before concatenating them along the channel dimension to obtain $F_1, F_2 \in \mathbb{R}^{2C\times H\times W}$. This process not only allows us to capture the similarity between Q_{cat} and K_1, K_2 but also between K_{cat} and Q_1, Q_2. The complementary fusion of these two distinct interaction features enhances the model's ability to process bitemporal data

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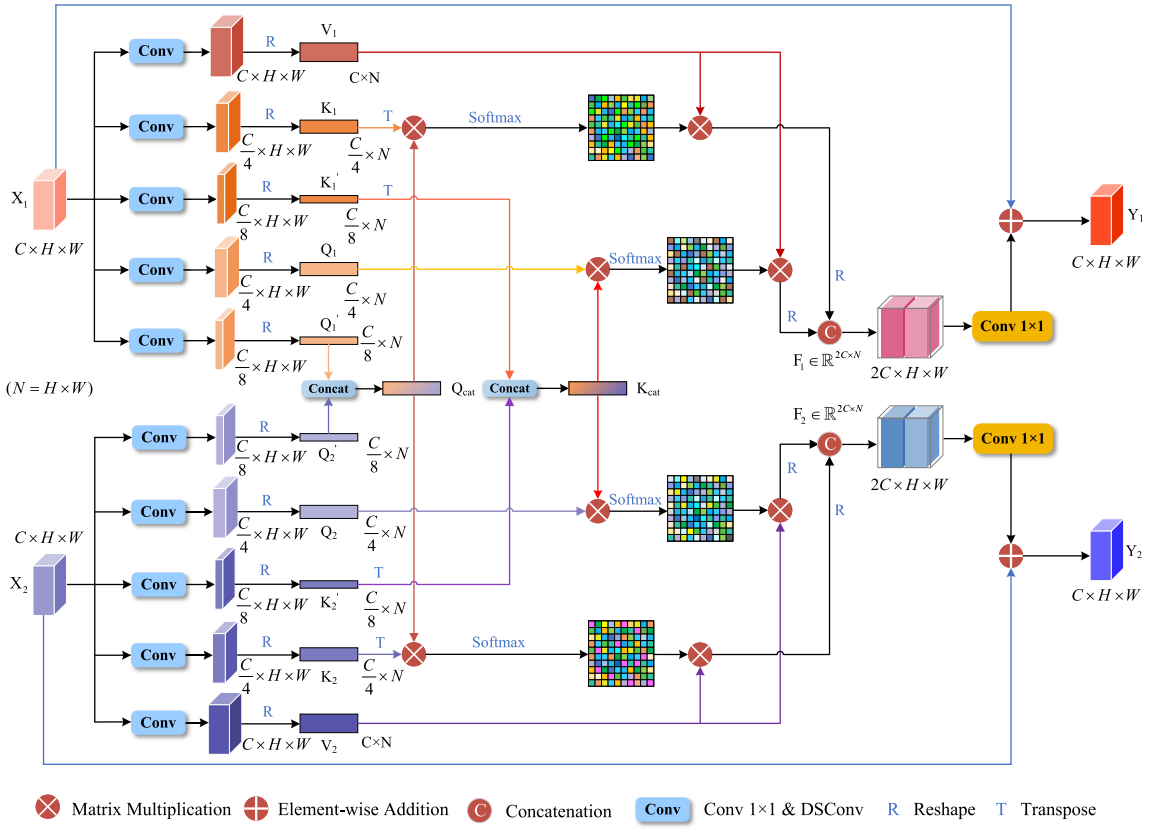


Fig. 3. Illustration of the proposed IAEM.

comprehensively in change detection tasks. This augmentation improves the model's perception and understanding of genuine change areas while enhancing its overall performance. The process can be formulated as follows:

$$F_1 = \text{Concat}(\text{Softmax}(\text{Concat}(Q_1', Q_2') \cdot K_1{}^T) \cdot V_1, \text{Softmax}(Q_1 \cdot \text{Concat}(K_1', K_2')^T) \cdot V_1) \quad (4)$$

$$F_2 = \text{Concat}(\text{Softmax}(\text{Concat}(Q_1', Q_2') \cdot K_2{}^T) \cdot V_2, \text{Softmax}(Q_2 \cdot \text{Concat}(K_1', K_2')^T) \cdot V_2) \quad (5)$$

where Softmax denotes the activation function, T denotes the transpose operation, and Concat$(\cdot)$ denotes a feature concatenation operation. By introducing more interactive information, we aim to effectively unify self-attention and cross-attention mechanisms. Subsequently, F_1 and F_2 are individually subjected to a 1×1 convolution to reduce the number of channels. Following this, they are further combined with the original features X_1 and X_2 via residual connections, resulting in Y_1 and Y_2. This enables the features of one image to more extensively focus on the characteristics of another image while retaining attention to its own features. This robust interaction between bitemporal features stimulates comprehensive interaction, significantly enhancing attention distribution toward genuine changing areas and suppressing irrelevant noise interference. The process can be formulated as follows:

$$Y_1 = \text{Conv}_{1\times1}(F_1) + X_1 \quad (6)$$

$$Y_2 = \text{Conv}_{1\times1}(F_2) + X_2 \quad (7)$$

where $\text{Conv}_{1\times1}(\cdot)$ represents a 1×1 convolutional layer with a batch normalization and a ReLU activation function.

D. Supervised Attention Module

The feature pyramid network [64] provides a classic architecture for multilevel feature fusion, progressing from coarse to fine. However, high-level features typically accompany more abstract semantic information and lack contextual guidance, while low-level features are closer to the original input data, containing more spatial and local information, often with greater noise. Therefore, achieving smooth and effective fusion of high-level and low-level features is crucial, as it can effectively reduce the loss of edge detail information and enhance model performance. Attention mechanisms, widely utilized to enhance model performance [65], [66], [67], guide models on where to focus. Inspired by the convolutional block attention module (CBAM) [65] and coordinate attention [68], we introduce a lightweight and more effective attention mechanism, named SAM, in the decoding stage. SAM recalibrates diverse features to achieve better fusion of multilevel differences. As depicted in Fig. 4, we use 1×1 convolutions, DSConvs [63], and our designed coordinate channel attention (CCA) to construct SAM.

CCA embeds positional information into channelwise and spatialwise attention, aggregating features along two spatial directions (W and H). This captures distant dependencies in one spatial direction while preserv-

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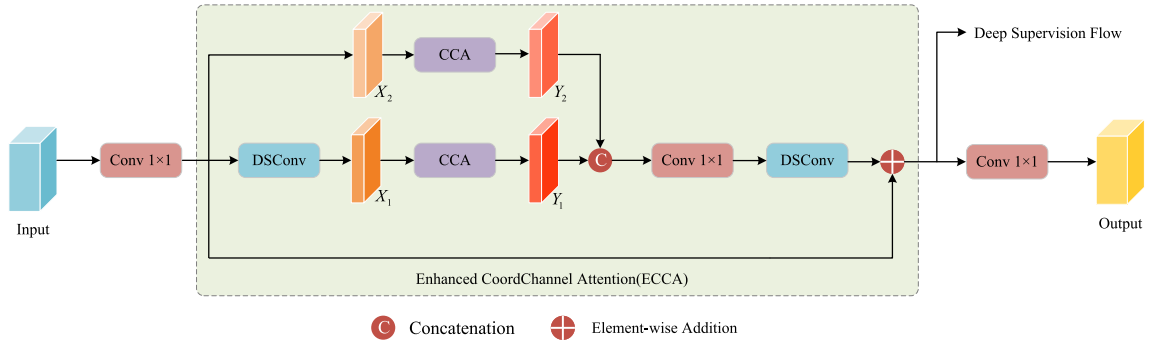


Fig. 4. Illustration of the proposed SAM.

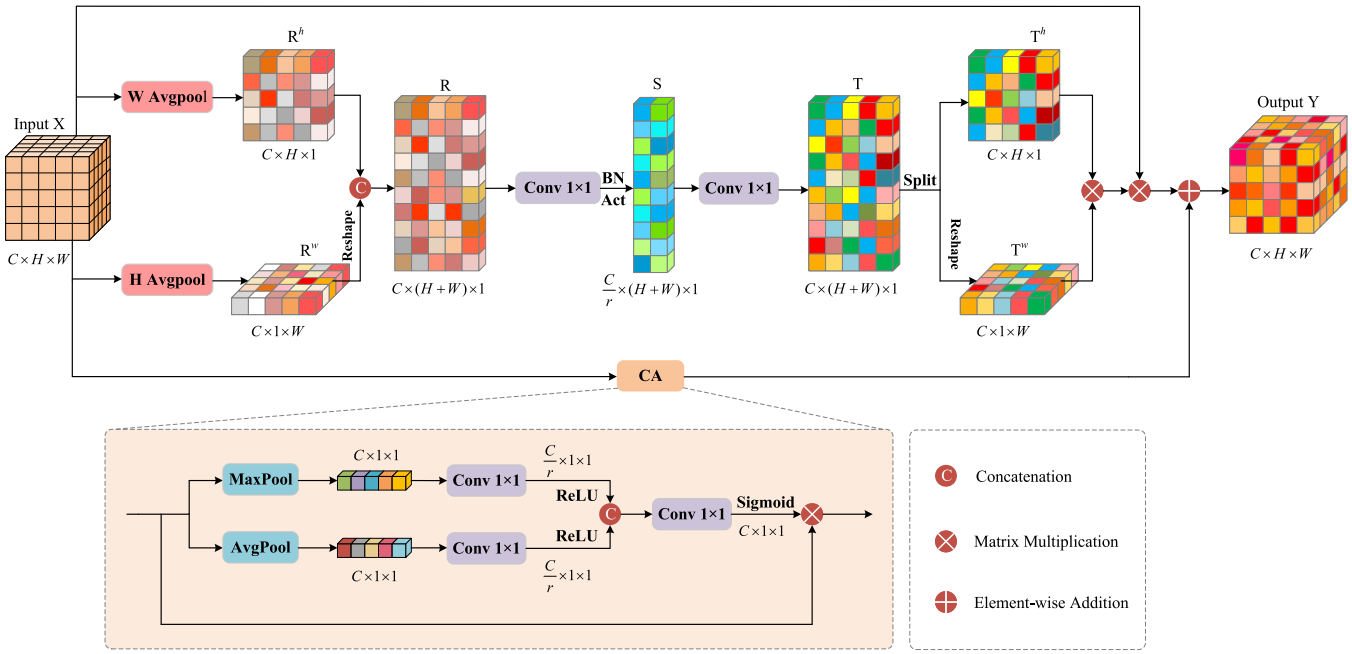


Fig. 5. Illustration of the proposed CCA.

ing precise positional information, strengthening the network's focus on the objects of interest and enhancing attention to the edges of changing objects. Initially, the features undergo a refinement extraction through a 1×1 convolution before entering the enhanced coordinate channel attention (ECCA) block. ECCA takes the 1×1 convolved features and DSConv-processed features as inputs to derive Y_1 and Y_2 separately within CCA. These outputs are concatenated along the channel dimension, forming a more enriched feature map. Subsequently, a 1×1 convolution reduces the channel dimensionality, followed by DSConv for further feature optimization. A residual connection combines this with the 1×1 convolved features, further enhancing the features. Finally, a 1×1 convolution generates the output. ECCA effectively readjusts the differential information from the encoding stage, minimizing parameters and computational overhead. By introducing additional supervision signals, gradients propagate better across different layers of the network. Moreover, ECCA not only leverages distant dependencies but also exploits channel and spatial positional information, enabling the network to precisely identify boundary changes in the region of interest.

Fig. 5 illustrates the specific framework of CCA. First, the input features are denoted as $X = [x_1, x_2, \cdots, x_C] \in \mathbb{R}^{C \times H \times W}$. CCA conducts adaptive average pooling along the W and H directions, generating two direction-aware feature maps. These are then concatenated along the spatial dimension. The process can be formulated as follows:

$$R = [W_{\text{avg}}(X); H_{\text{avg}}(X)^T] \tag{8}$$

where $W_{\text{avg}}(\cdot)$ denotes adaptive average pooling along the W direction, $H_{\text{avg}}(\cdot)$ signifies adaptive average pooling along the W direction, ${H_{\text{avg}}}^T(\cdot)$ indicates transpose $H_{\text{avg}}(\cdot)$ for subsequent concatenation, and [;] represents spatial dimension concatenation. Then, it undergoes channel dimension transformation via a 1×1 convolution, denoted as

$$T = f^{1\times1}(\delta(f^{1\times1}(R))) \tag{9}$$

where $f^{1\times1}(\cdot)$ denotes a 2-D convolution with a kernel size of 1, $\delta(\cdot)$ signifies the implementation of batch normalization,

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and ReLU serves as the activation function, $T \in \mathbb{R}^{C\times(H+W)\times 1}$. Then, CCA splits T into two separate tensors, $T^h \in \mathbb{R}^{C\times H\times 1}$ and $T^w \in \mathbb{R}^{C\times 1\times W}$, along the spatial dimension, resulting in the final output feature $Y = [y_1, y_2, \cdots, y_C] \in \mathbb{R}^{C\times H\times W}$, denoted as

$$Y = T^h \times T^w \times X + \mathrm{CA}(X) \tag{10}$$

where CA(·) denotes the channel attention operation, as illustrated in Fig. 5, which employs global average pooling and global max pooling for feature extraction. Subsequently, channel transformations are individually performed using 1 × 1 convolutions. Due to the relatively large number of channels during downsampling, we have set a channel reduction factor r in the CA module to decrease the channel count. Through multiple experiments, we found that setting r to 4 yields the best performance. The CA module can be formulated as follows:

$$y = \mathrm{Sigmoid}(\mathrm{Conv}_{1\times1}(\mathrm{Concat}(\mathrm{Conv}_{1\times1}(\max(x)) \\ \mathrm{Conv}_{1\times1}(\mathrm{avg}(x))))) \times x \tag{11}$$

where $\mathrm{Conv}_{1\times1}(\cdot)$ represents a 1 × 1 convolutional layer with a batch normalization and a ReLU activation function, max(·) denotes the global maximum pooling operation, avg(·) represents the global average pooling, Concat(·) denotes a feature concatenation operation, and Sigmoid(·) denotes the activation function.

CCA encodes both channel and spatial information, applying horizontal and vertical attention maps to input features. This enhances spatial integration and object localization, producing more distinguishable features and improving edge detection of changing targets.

E. Loss Function

For complex deep CNN models, relying solely on the loss function at the final output layer for backpropagation can lead to gradient vanishing as it propagates to deeper layers, destabilizing parameter optimization in intermediate layers and potentially harming overall performance. Additionally, in change detection tasks, the proportion of changed regions is typically much smaller than that of unchanged regions, causing a class imbalance issue. To address these challenges, we used BCEWithLogitsLoss as the loss function and applied a deep supervision strategy using SAM at intermediate layers during decoding. By setting weight coefficients within the loss function, we focused the training more on the changed regions. As a result, the total loss during training consisted of the main branch loss and three auxiliary losses. The BCE loss function can be expressed as follows:

$$L_{\mathrm{BCE}}(c, g) = g \cdot \log c + (1 - g) \cdot \log(1 - c) \tag{12}$$

where · denotes the dot product operation, and c and g represent the predicted change map and the corresponding ground-truth label map, respectively. The total training loss is represented as follows:

$$L_{\mathrm{total}} = \lambda_1 L_{\mathrm{BCE}}(c_1, g) + \lambda_2 L_{\mathrm{BCE}}(c_2, g) \\ + \lambda_3 L_{\mathrm{BCE}}(c_3, g) + \lambda_4 L_{\mathrm{BCE}}(c_4, g) \tag{13}$$

where λ_1 denotes the weight coefficient for the primary branch loss, and λ_2, λ_3, and λ_4 are the weight coefficients for the three auxiliary losses, respectively. c_1 denotes the predicted change map from the primary branch, and c_2, c_3, and c_4 are the predicted change maps from the three auxiliary branches, respectively.

IV. Experiments

A. Datasets

We selected the LEVIR-CD [69], Guangzhou dataset (GZ-CD) [70], and Sun Yat-sen University dataset (SYSU-CD) [48] for our experiments due to their significant differences in resolution, scene complexity, time span, and data scale, which comprehensively validate the proposed method's generalization and robustness. The LEVIR-CD dataset contains images with notable land use changes, facilitating the evaluation of long-term change detection performance. The GZ-CD dataset, covering seasonal changes in Guangzhou, is suitable for assessing performance in small-sample scenarios. The SYSU-CD dataset, with its large-scale and diverse complex scenes, tests adaptability in challenging change detection environments. Thus, the combined use of these datasets ensures the effectiveness of the proposed method under various conditions.

1) LEVIR-CD: The LEVIR-CD consists of 637 pairs of high-resolution remote sensing images, with a resolution of 0.5 m/pixel, each pair being composed of ultrahigh-resolution Google Earth images sized at 1024 × 1024 pixels. These image pairs exhibit a time span between 5 and 14 years and encompass substantial land use changes, particularly notable in areas featuring villas, apartments, large warehouses, garages, and more. These images originate from 20 different regions across various cities in TX, USA. Due to GPU memory constraints, each image in the dataset was cropped into 16 nonoverlapping patches of size 256 × 256 pixels. The LEVIR-CD dataset was further divided into sets of 7120/1024/2048 pairs in a ratio of 7:1:2 for training/validation/testing, respectively.

2) GZ-CD: The GZ-CD comprises VHR satellite imagery CD dataset with a spatial resolution of 0.55 m/pixel, covering the suburban areas of Guangzhou, China, spanning from 2006 to 2019. Obtained through Google Earth services, a total of 19 pairs of seasonal change images were collected, ranging in spatial dimensions from 1006 × 1168 to 4936 × 5224. We uniformly segmented the images into sizes of 256 × 256. These image blocks were further divided into sets of 2504/313/313 pairs in an 8:1:1 ratio for training/validation/testing, respectively. As such, GZ-CD represents a relatively smaller dataset.

3) SYSU-CD: The SYSU-CD comprises 20 000 pairs of bitemporal remote sensing image blocks, sized at 256 × 256 with a spatial resolution of 0.5 m. The SYSU-CD dataset is further divided into sets of 12000/4000/4000 pairs in a 6:2:2 ratio for training/validation/testing, respectively. This dataset encompasses various complex change scenarios, including road expansion, new urban construction, vegetation changes, suburban expansion, and groundwork before construction.

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B. Implementation Details

1) Experimental Environment and Parameter Settings: All experiments in this article were conducted on a GeForce RTX 3080 based on PyTorch, using Python 3.8 as the programming language. Prior to model training, we performed data augmentation on input images, including cropping, resizing, and rotation. We utilized BCEWithLogitsLoss as the model's loss function and employed Adam as the optimizer during model training. Given the effectiveness of dynamically adjusting the learning rate, we adopted the Poly learning rate strategy, where the learning rate for each epoch was adjusted to $\mathrm{lr} \times (1 - \mathrm{epoch}/\mathrm{max_epoch})^{\mathrm{power}}$. The batch size was set to 8, the training epoch (max_epoch) was set to 200, and the initial learning rate (lr) was set at 0.0001 with a power of 0.9. During training, we employed deep supervision on the network, consisting of four parts in the loss function: the main branch loss and three auxiliary losses, where the weight coefficients λ_1, λ_2, λ_3, and λ_4 were all set to 1.

2) Evaluation Metrics: In addition to intuitively comparing the goodness of results in change detection, it is also important to calculate numerical metrics between the predicted result map and the true label map to quantitatively analyze the strengths and weaknesses of the algorithm. To provide a more comprehensive evaluation of the algorithm's performance, we introduce six metrics, including precision (PR), false negative rate (FNR), F1-score (F1), intersection over union (IoU), error rate (ER), and Kappa. These evaluation metrics include both positive and negative indicators, and are calculated as follows:

$$\mathrm{PR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{14}$$

$$\mathrm{FNR} = \frac{\mathrm{FN}}{\mathrm{TP} + \mathrm{FN}} \tag{15}$$

$$\mathrm{F1} = \frac{2\mathrm{PR} \cdot \mathrm{RC}}{\mathrm{PR} + \mathrm{RC}} \tag{16}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{17}$$

$$\mathrm{ER} = \frac{\mathrm{FP} + \mathrm{FN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{18}$$

$$\mathrm{P}_e = \frac{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN}) + (\mathrm{FN} + \mathrm{TN})(\mathrm{TP} + \mathrm{TN})}{(\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN})^2} \tag{19}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA} - \mathrm{P}_e}{1 - \mathrm{P}_e} \tag{20}$$

where TP stands for true positive, referring to the correctly predicted change regions. FP represents false positive, indicating the incorrectly predicted change regions. TN stands for true negative, representing correctly predicted unchanged regions. FN refers to false negative, indicating the incorrectly predicted unchanged regions.

3) Comparative Methods: To evaluate the effectiveness and superiority of our proposed ISDANet, we compared it against several state-of-the-art (SOTA) CD methods, including FC-EF [71], FC-Siam-Diff [71], FC-Siam-Conc [71], ChangeNet [72], deeply supervised image fusion network (DSIFN) [73], bitemporal image transformer (BIT) network [36], densely connected Siamese network (SNUNet) [74], intrascale cross-interaction and interscale feature fusion network (ICIFNet) [75], dual-branch multilevel intertemporal network (DMINet) [38], attention-guided Siamese network (SAGNet) [41], spatial focused bitemporal interactive network (SFBI-Net) [76], and spatiotemporal enhancement and interlevel fusion network (SEIFNet) [77].

All competing methods were trained from scratch. Considering the significance of hyperparameters on model performance, to ensure fair comparison experiments, we followed the settings as reported in their original papers.

TABLE I

QUANTITATIVE RESULTS OF VARIOUS CHANGE DETECTION METHODS ON THE LEVIR-CD DATASET. (BEST VALUES ARE HIGHLIGHTED IN BOLD)

Methods	PR(%)	Kappa(%)	IoU(%)	F1(%)	FNR(%)	ER(%)
FC-EF	85.58	82.30	71.19	83.17	19.11	1.67
FC-Siam-Diff	89.49	84.08	73.69	84.85	19.33	1.47
FC-Siam-Conc	86.76	85.56	75.89	86.29	14.17	1.39
ChangeNet	91.63	88.63	80.49	89.19	13.12	1.07
DSIFN	91.53	87.75	79.12	88.34	14.30	1.13
BIT	91.26	89.33	81.59	89.86	11.49	1.02
SNUNet	91.51	89.46	81.79	89.98	11.49	1.00
ICIFNet	91.31	89.16	81.24	89.18	12.77	1.44
DMINet	92.02	89.31	81.56	89.85	12.27	1.21
SAGNet	91.79	89.58	81.98	90.10	11.24	0.98
SFBI-Net	91.85	89.45	83.05	90.65	10.54	1.08
SEIFNet	91.56	89.30	83.22	90.78	10.65	1.79
Ours	**92.19**	**90.59**	**83.63**	**91.06**	**9.92**	**0.90**

C. Comparative Study

We conducted comparisons with the aforementioned 12 SOTA change detection methods on the LEVIR-CD, GZ-CD, and SYSU-CD datasets, followed by quantitative and qualitative analyses of the experimental results. Additionally, to ensure the accuracy of each method's results and fairness in the comparative experiments, we utilized the same datasets for training and testing across all methods. The evaluation results of each method on the respective datasets are presented below.

1) Comparative Experiments on LEVIR-CD: The quantitative results of various methods on the LEVIR-CD dataset are shown in Table I. It can be observed from Table I that our method demonstrates superior performance across several metrics when compared to other methods. Specifically, the precision, F1, and IoU values are 92.19%, 91.06%, and 83.63%, respectively. Compared to the best-performing competing methods, our approach shows an improvement of 0.28% in F1 and 0.41% in IoU. Visually, as depicted in Fig. 6, our network on the LEVIR-CD dataset exhibits higher F1 scores and lower standard deviations compared to several selected competitors. Clearly, our ISDANet outperforms others.

Our method's visual results alongside other methods on the LEVIR-CD dataset are depicted in Fig. 7. From the illustration, it is evident that our approach adeptly identifies variations in both dense and sporadic small-scale structures, as well as substantial buildings. In cases of continuous small-scale structure changes, competing methods often demonstrate jagged or

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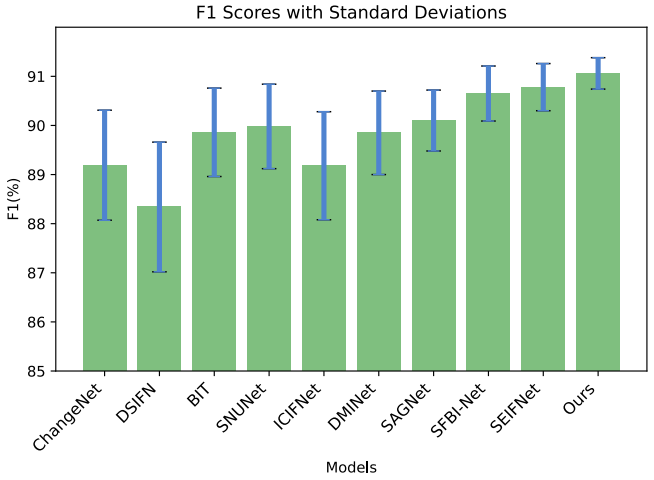


Fig. 6. Comparison on the LEVIR-CD dataset. The bar graph represents the F1 value, and the standard deviation is displayed at the top.

clustered predicted boundaries, leading to false positives and negatives. Conversely, our method offers clearer segmentation of change boundaries, notably distinguishing between adjacent buildings. For fragmented structural changes, other methods exhibit higher rates of false positives and negatives, especially in detecting changes among surrounding elements like trees and roads, where our method performs better. In the context of significant structural changes, while other methods exhibit less smoothness overall and have difficulty detecting boundaries in change areas, often exhibiting substantial missed areas, our method, while also having some missed areas, yields the best overall change prediction, particularly demonstrating efficient detection of larger structural changes. This aligns with the quantitative results in Table I. Both in quantitative metrics and qualitative analysis, our ISDANet outperforms recent SAGNet, SFBI-Net, and SEIFNet releases in terms of performance.

2) Comparative Experiments on GZ-CD: The quantitative results of various methods on the GZ-CD dataset are presented in Table II. From the table, it is evident that our method achieved the best performance across several metrics when compared to other methods, with precision, F1, and IoU values of 89.72%, 88.25%, and 78.97%, respectively. Compared to the best-performing competitive method, our approach exhibited an improvement of 1.6% in F1 and 2.74% in IoU. Visually, Fig. 8 displays the F1 values and their standard deviation of our network on GZ-CD against selected competitive counterparts. Clearly, our ISDANet outperforms others.

The visual results of our method and other approaches on the GZ-CD dataset are shown in Fig. 9. From the illustration, it can be observed that for relatively dense and regular changes in large factories, other algorithms tend to generate significant areas of false positives and false negatives, while our algorithm performs the best in this regard. Additionally, as depicted in Fig. 9, other methods exhibit substantial false negatives when faced with elongated and small-target building changes. Furthermore, when distinguishing these building changes, especially at the edges of changed regions, other methods tend to have more false positives. In contrast, our method provides clearer segmentation of changed target edges and more accurate detection of small targets, displaying optimal visual behavior on GZ-CD, consistent with the quantitative results in Table II. Whether in quantitative results or qualitative analysis, our ISDANet outperforms the latest SAGNet, SFBI-Net, and SEIFNet.

TABLE II

QUANTITATIVE RESULTS OF VARIOUS CHANGE DETECTION METHODS ON THE GZ-CD DATASET. (BEST VALUES ARE HIGHLIGHTED IN BOLD)

Methods	PR(%)	Kappa(%)	IoU(%)	F1(%)	FNR(%)	ER(%)
FC-EF	79.86	69.44	56.24	71.99	34.47	4.72
FC-Siam-Diff	82.70	65.55	51.72	68.18	42.01	5.01
FC-Siam-Conc	82.16	68.67	55.26	71.19	37.20	4.71
ChangeNet	88.63	84.32	75.01	85.72	17.01	2.56
DSIFN	89.48	79.83	68.76	81.49	24.54	3.09
BIT	86.80	82.80	72.94	84.35	17.96	2.82
SNUNet	89.00	85.54	76.75	86.85	15.20	2.38
ICIFNet	88.09	83.05	73.25	84.56	18.69	2.75
DMINet	86.62	83.17	73.45	84.70	17.15	2.77
SAGNet	89.56	84.98	75.91	86.30	15.95	2.42
SFBI-Net	89.03	85.35	76.05	86.36	15.38	2.62
SEIFNet	88.98	85.62	76.23	86.65	15.12	2.57
Ours	**89.72**	**87.07**	**78.97**	**88.25**	**13.17**	**2.14**

TABLE III

QUANTITATIVE RESULTS OF VARIOUS CHANGE DETECTION METHODS ON THE SYSU-CD DATASET. (BEST VALUES ARE HIGHLIGHTED IN BOLD)

Methods	PR(%)	Kappa(%)	IoU(%)	F1(%)	FNR(%)	ER(%)
FC-EF	78.78	70.97	63.56	77.72	23.31	10.37
FC-Siam-Diff	80.35	64.42	55.11	71.06	25.74	11.29
FC-Siam-Conc	81.51	71.80	64.17	78.18	24.89	9.89
ChangeNet	79.91	68.19	60.33	75.25	28.89	11.03
DSIFN	78.82	73.76	66.72	80.04	18.70	9.56
BIT	81.22	70.81	63.09	77.37	26.13	10.19
SNUNet	79.37	72.42	65.13	78.88	21.61	9.90
ICIFNet	78.23	69.17	61.62	76.25	25.62	10.92
DMINet	81.54	74.59	67.06	80.28	20.56	8.85
SAGNet	81.25	76.57	69.31	81.87	18.24	8.28
SFBI-Net	80.85	76.50	69.56	82.03	18.64	8.39
SEIFNet	80.52	76.34	69.72	82.21	20.14	8.32
Ours	**82.06**	**77.60**	**70.90**	**82.97**	**16.62**	**8.17**

3) Comparative Experiments on SYSU-CD: The quantitative results of various methods on the SYSU-CD dataset are presented in Table III. From Table III, it can be observed that our method outperforms others in several metrics, with precision, F1, and IoU values of 82.06%, 82.97%, and 70.90%, respectively. In comparison to the best-performing competing method, our approach has improved by 0.76% and 1.18% in terms of F1 and IoU, respectively. Visually, Fig. 10 displays the F1 values and their standard deviation of our network and several selected competitors on SYSU-CD. Clearly, the performance of our ISDANet surpasses the others.

The visual results of our method and other approaches on the SYSU-CD dataset are illustrated in Fig. 11. From the illustration, it can be observed that whether it is for detecting changes in buildings, vegetation, or farmland, other algorithms tend to generate significant areas of false positives and

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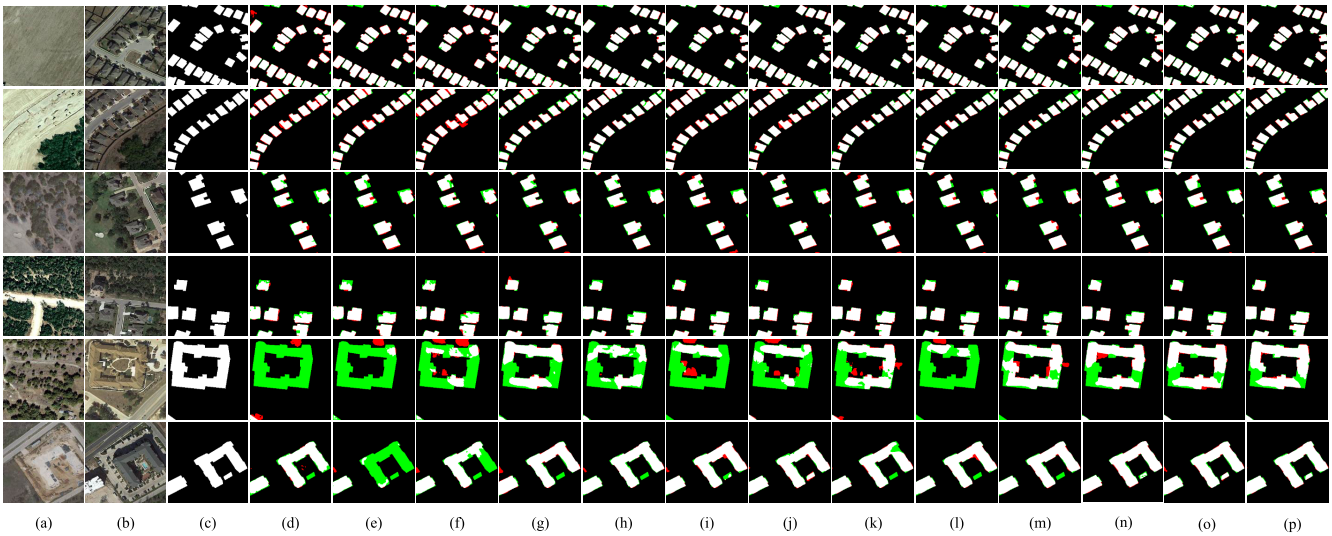


Fig. 7. Visualization results of different change detection methods on LEVIR-CD dataset. (a) and (b) Remote sensing images before and after the change. (c) Ground truth. (d) FC-EF. (e) FC-Siam-Diff. (f) FC-Siam-Conc. (g) ChangeNet. (h) DSIFN. (i) BIT. (j) SNUNet. (k) ICIFNet. (l) DMINet. (m) SAGNet. (n) SFBI-Net. (o) SEIFNet. (p) Ours. In the change map, white pixels represent TP, black pixels represent TN, red pixels represent FP, and green pixels represent FN.

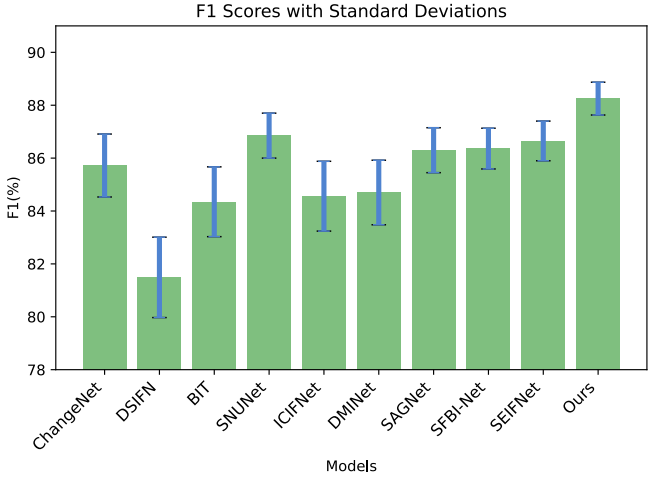


Fig. 8. Comparison on the GZ-CD dataset. The bar graph represents the F1 value, and the standard deviation is displayed at the top.

false negatives, whereas our algorithm excels in this regard. Additionally, when changes occur due to the appearance of ships on water bodies, other methods display clear deficiencies in detecting ship edge regions, resulting in numerous false positives and negatives. Particularly, other methods perform poorly when distinguishing concentrated ships and nearby small target changes in areas affected by optical refraction on water bodies, exhibiting numerous false negatives. Overall, our method provides clearer segmentation of changed target edges and more accurate detection of small targets in vegetation, buildings, and water bodies, demonstrating optimal visual behavior on SYSU-CD, consistent with the quantitative results in Table III. Whether through quantitative results or qualitative analysis, our ISDANet outperforms the latest SAGNet, SFBI-Net, and SEIFNet.

D. Model Efficiency Analysis

In addition to the quantitative and qualitative analyses of all methods, we further evaluated the efficiency of various approaches using the number of parameters (Params) and floating-point operations (FLOPs), further illustrating the effectiveness of our method. The image size used for efficiency comparison was 256 × 256 pixels.

TABLE IV
EFFICIENCY COMPARISON OF DIFFERENT METHODS

Methods	Params (M)	FLOPs (G)	Time (ms)
FC-EF	1.35	3.57	5.13
FC-Siam-Diff	1.35	4.72	7.59
FC-Siam-Conc	1.55	5.32	7.14
ChangeNet	47.20	10.91	18.28
DSIFN	35.73	82.26	11.81
BIT	3.49	10.63	17.32
SNUNet	12.03	54.82	9.58
ICIFNet	23.84	24.51	50.63
DMINet	6.24	14.55	15.29
SAGNet	32.23	12.25	23.32
SFBI-Net	19.59	35.24	32.82
SEIFNet	27.91	8.37	28.65
Ours	6.93	3.46	21.01

Table IV presents the Params, FLOPs, and time costs for all methods. The first three networks have smaller Params, FLOPs, and time costs but exhibit poor F1 scores, indicating suboptimal performance in CD tasks. ChangeNet has the largest Params, followed by DSIFN, but their performance is still lacking. SNUNet has moderate Params and FLOPs with average performance. While ICIFNet and SAGNet show good performance, their large Params and FLOPs require more computational resources. SFBI-Net and SEIFNet also perform well but have high Params and FLOPs. Our network slightly exceeds BIT in Params but significantly outperforms it. It is comparable to DMINet in Params yet performs better. With the lowest FLOPs and best results, our method strikes a balance among model complexity, time cost, and accuracy. In terms of time costs, it is mid-range. Scatter plots of F1 scores

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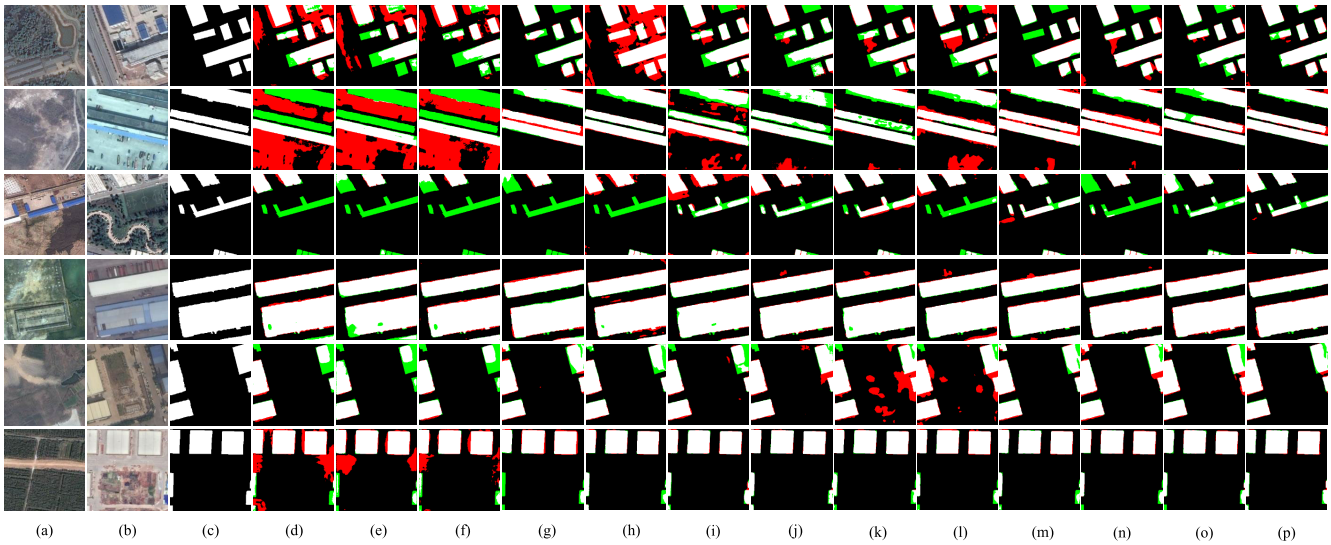


Fig. 9. Visualization results of different change detection methods on GZ-CD dataset. (a) and (b) Remote sensing images before and after the change. (c) Ground truth. (d) FC-EF. (e) FC-Siam-Diff. (f) FC-Siam-Conc. (g) ChangeNet. (h) DSIFN. (i) BIT. (j) SNUNet. (k) ICIFNet. (l) DMINet. (m) SAGNet. (n) SFBI-Net. (o) SEIFNet. (p) Ours. In the change map, white pixels represent TP, black pixels represent TN, red pixels represent FP, and green pixels represent FN.

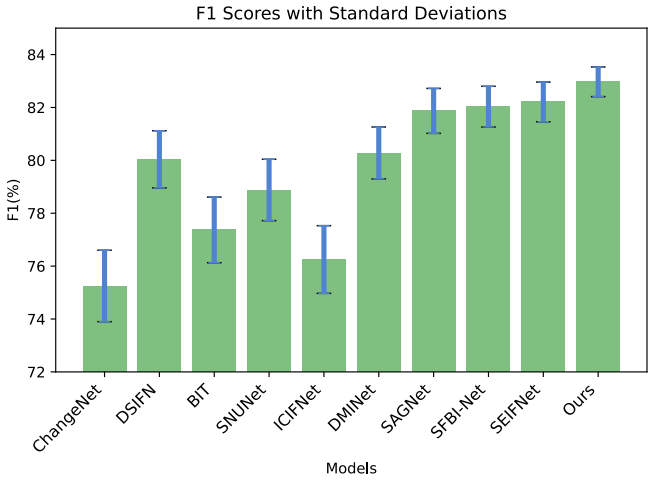


Fig. 10. Comparison on the SYSU-CD dataset. The bar graph represents the F1 value, and the standard deviation is displayed at the top.

versus Params and FLOPs across three datasets, shown in Fig. 12, further highlight that our method achieves the best performance with fewer Params and lower computational costs on all datasets.

E. Ablation Study

In order to comprehensively evaluate the effectiveness of each module within our network, we conducted ablation experiments across three datasets. In these experiments, the training strategies for all models remained consistent, systematically adding proposed modules onto the backbone network to assess their individual contributions. Table V presents the results of the ablation experiments, with a primary focus on the IoU and F1 metrics to validate our module's impact.

NFAM: Ablation experiments were conducted on our NFAM, which integrates features from adjacent scales of the backbone network, encompassing both high-level semantic and low-level semantic features. This integration further

TABLE V

QUANTITATIVE PERFORMANCE COMPARISON OF ABLATION STUDY RESULTS ACROSS DIFFERENT DATASETS. (BEST VALUES ARE HIGHLIGHTED IN BOLD)

Dataset	Methods	NFAM	IAEM	SAM	IoU	F1
LEVIR-CD	Baseline+				80.31	89.08
	Baseline+	✓			81.36	89.72
	Baseline+	✓	✓		82.40	90.35
	Baseline+	✓	✓	✓	**83.63**	**91.06**
GZ-CD	Baseline+				73.94	85.02
	Baseline+	✓			76.41	86.63
	Baseline+	✓	✓		77.59	87.38
	Baseline+	✓	✓	✓	**78.97**	**88.25**
SYSU-CD	Baseline+				67.65	80.70
	Baseline+	✓			68.86	81.38
	Baseline+	✓	✓		69.67	82.12
	Baseline+	✓	✓	✓	**70.90**	**82.97**

enhances scene information from adjacent-scale features, augmenting the representation capability of bitemporal features. As observed from Table V, incorporating NFAM onto the backbone network resulted in improvements of 1.05%/0.64%, 2.47%/1.61%, and 1.21%/0.68% for IoU and F1 on the three datasets, respectively.

IAEM: We conducted ablation experiments on our IAEM to illustrate its effectiveness. The IAEM was devised to enhance the interaction and coupling of bitemporal features, thereby strengthening the attention distribution over genuine change regions. As shown in Table V, incorporating IAEM onto the backbone network after NFAM resulted in respective increments of 1.04%/0.63%, 1.18%/0.75%, and 0.81%/0.74% for IoU and F1 on the three datasets. However, traditional self-attention mechanisms focus on globally attending to individual images by dynamically measuring similarity between representation vectors. Feng et al. [38] proposed JointAtt,

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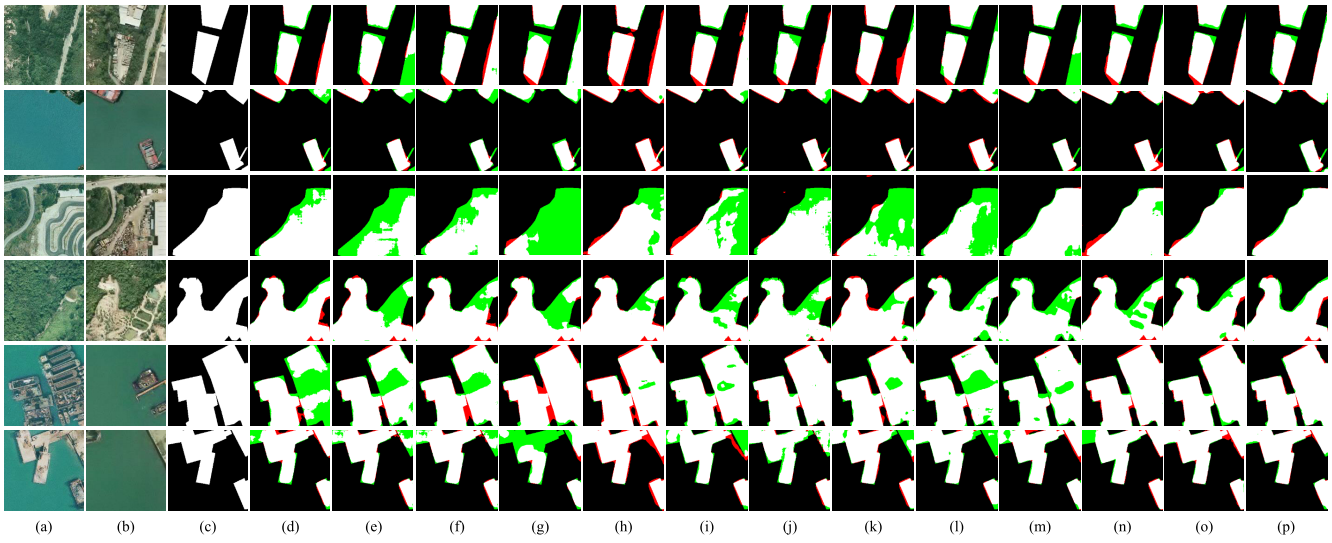


Fig. 11. Visualization results of different change detection methods on SYSU-CD dataset. (a) and (b) Remote sensing images before and after the change. (c) Ground truth. (d) FC-EF. (e) FC-Siam-Diff. (f) FC-Siam-Conc. (g) ChangeNet. (h) DSIFN. (i) BIT. (j) SNUNet. (k) ICIFNet. (l) DMINet. (m) SAGNet. (n) SFBI-Net. (o) SEIFNet. (p) Ours. In the change map, white pixels represent TP, black pixels represent TN, red pixels represent FP, and green pixels represent FN.

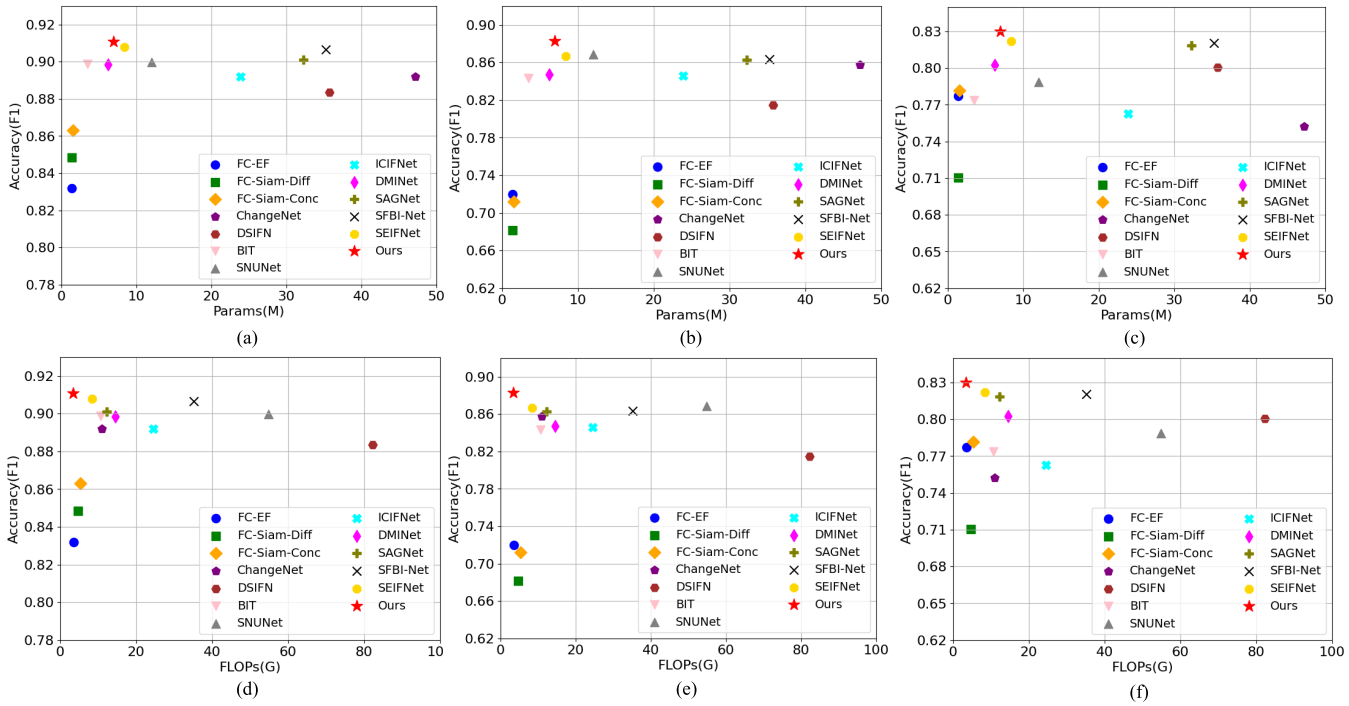


Fig. 12. Model complexity comparison of different methods in terms of Params (memory cost), FLOPs (computational cost), and F1-score on the LEVIR-CD, GZ-CD, and SYSU-CD datasets, respectively. (a) and (d) LEVIR-CD dataset. (b) and (e) GZ-CD dataset. (c) and (f) SYSU-CD dataset. Our proposed ISDANet method shows superior accuracy with fewer parameters and lower computation costs.

attempting to combine self-attention and cross-attention by exchanging query vectors Q, yet this interaction between bitemporal features is not comprehensive enough, potentially overlooking significant change information. Especially with this insufficient interaction, detailed information at the edges of real change regions and features of small targets are likely to be lost. In contrast, our designed IAEM not only exchanges query vectors Q but also exchanges key vectors K, resulting in a more enriched interactive feature. Table VI presents the ablation results on the three datasets, focusing solely on F1. From Table VI, it is evident that models using IAEM achieve the highest F1 scores, indicating superior performance and highlighting the capability of IAEM in better coupling bitemporal features.

SAM: We also conducted ablation experiments on SAM. Our SAM effectively integrates multilevel differential features and provides additional depth supervision signals to enhance network convergence. As shown in Table V, incorporating

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TABLE VI
ABLATION STUDY ON DIFFERENT ATTENTION MECHANISMS

Methods	LEVIR-CD	GZ-CD	SYSU-CD
Without	89.78	86.89	81.64
Self-Att	89.91	87.38	81.95
Cross-Att	89.95	87.43	82.01
Joint-Att	90.16	87.62	82.26
IAEM	**91.06**	**88.25**	**82.97**

TABLE VII
ABLATION STUDY ON DIFFERENT ATTENTION MECHANISMS

Methods	LEVIR-CD	GZ-CD	SYSU-CD
Without	90.35	87.38	82.12
CBAM	90.43	87.46	82.26
CCA	90.69	87.65	82.47
SAM	**91.06**	**88.25**	**82.97**

SAM resulted in IoU and F1 improvements of 1.23%/0.71%, 1.20%/0.87%, and 1.23%/0.85% across the three datasets. Traditional CBAM combines CA and spatial attention in a simplistic manner, lacking precise localization for edge details of changing objects and small targets. Therefore, we designed the CCA module to enhance feature representation capability, integrating it into our SAM. Table VII illustrates the necessity of introducing attention mechanisms during the decoding phase to progressively fuse multilevel differential features, showcasing superior performance with our SAM. Fig. 13 visually demonstrates the effectiveness of our proposed SAM in capturing edge details of change regions and features of small targets. The results indicate that our method surpasses SEIFNet in boundary recognition and small target capture. Networks incorporating SAM notably outperform those without SAM, underscoring its effectiveness in progressively aggregating differential features from coarse to fine levels.

Effect Using Different Backbones: To evaluate the impact of different backbones on our model's performance, we conducted ablation experiments using ResNet18, ResNet34, ResNet50, ShuffleNet [78], EfficientNet [79], and MobileNetV2. As shown in Table VIII, using ResNet18 resulted in slightly lower performance compared to MobileNetV2, while requiring more computational resources. Increasing the depth with ResNet34 and ResNet50 improved performance but at the cost of significantly higher resource consumption. ShuffleNet and EfficientNet, though lightweight, underperformed compared to MobileNetV2 and consumed more resources. Clearly, MobileNetV2 is the optimal backbone, achieving high performance with minimal computational cost.

Effect of Using Pretrained Weights: To verify the impact of pretrained weights on model performance, we conducted a set of experiments using both pretrained MobileNetV2 and MobileNetV2 trained from scratch for remote sensing image change detection. As shown in Table IX, the pretrained MobileNetV2 demonstrated significant performance advantages in this task. Although natural scenes and remote sensing scenes differ greatly in terms of data distribution, texture features, and object categories, the low-level features (such as edges and textures) and mid-level features (such as shapes and patterns) learned from large-scale datasets like ImageNet exhibit strong generalizability. These features provide a solid initial representation for remote sensing tasks, helping to accelerate the training process and effectively enhance model performance.

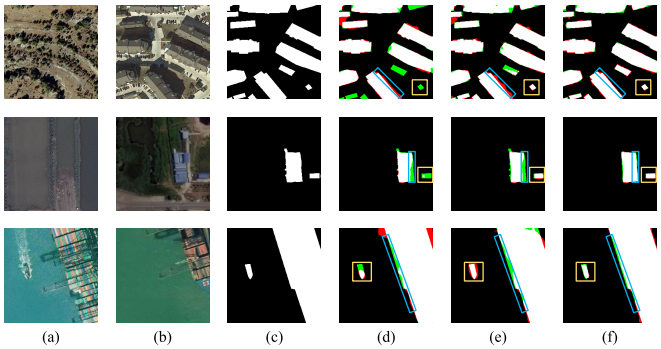


Fig. 13. (a) and (b) Remote sensing images before and after the change. (c) Ground truth. (d) Visualization results of SEIFNet. (e) Visualization results of our network without SAM. (f) Visualization results of our network with SAM. The samples are from the LEVIR-CD, GZ-CD, and SYSU-CD datasets, respectively.

TABLE VIII
ABLATION STUDY ON ISDANet WITH DIFFERENT BACKBONES

Backbones	Params (M)	FLOPs (G)	LEVIR-CD	GZ-CD	SYSU-CD
ResNet18	16.73	7.79	91.04	88.13	82.81
ResNet34	26.84	12.62	91.12	88.29	83.03
ResNet50	32.43	14.81	91.28	88.48	83.21
ShuffleNet	10.83	4.17	88.49	86.26	80.35
EfficientNet	46.50	20.20	89.31	87.12	81.23
MobileNetV2	6.93	3.46	91.06	88.25	82.97

TABLE IX
EXPERIMENT ON THE IMPACT OF USING PRETRAINED WEIGHTS ON MODEL PERFORMANCE (ISDANet* REPRESENTS ISDANet WITH MobileNetV2 TRAINED FROM SCRATCH AS THE BACKBONE, WHILE ISDANet REPRESENTS ISDANet WITH PRETRAINED MobileNetV2 AS THE BACKBONE)

Methods	LEVIR-CD	GZ-CD	SYSU-CD
ISDANet*	90.15	86.93	82.03
ISDANet	**91.06**	**88.25**	**82.97**

Effect of Different Weight Coefficients in the Loss Function: To select the four weight coefficients in the total loss function, we conducted multiple experiments to evaluate the impact of different weight combinations on model performance. The total loss function consists of a primary loss and auxiliary losses. Therefore, we fixed the primary loss weight coefficient λ_1 at 1 and varied the auxiliary loss weight coefficients $\lambda_2 = \lambda_3 = \lambda_4$ within the range [0, 1]. As shown in Table X, we found that setting all four weight coefficients to 1 achieved the best model performance. Specifically, this combination provided an optimal balance across various evaluation metrics. With this configuration, both the primary loss and auxiliary losses contributed equally during the training process. Our experiments demonstrated that increasing the proportion of

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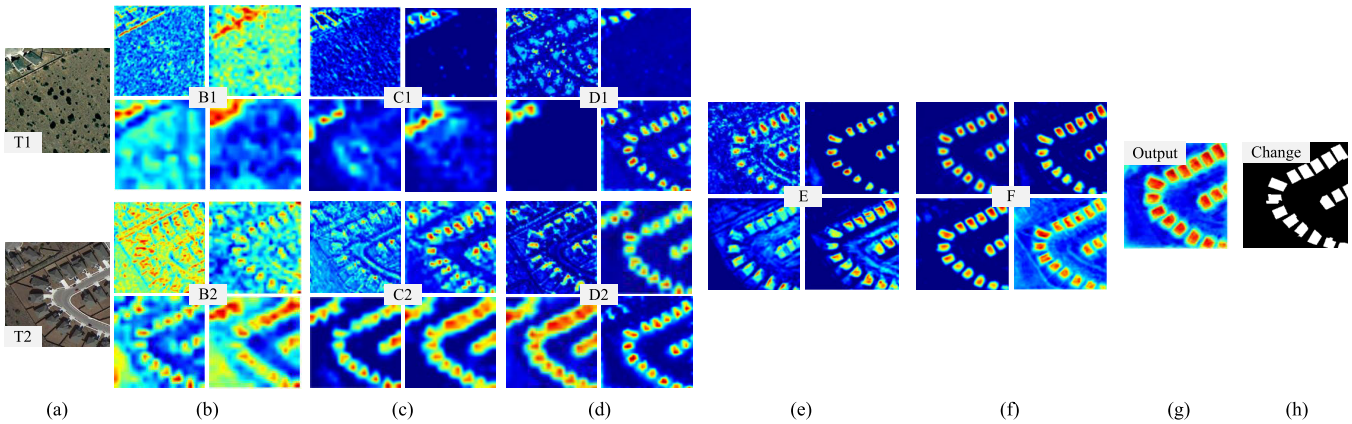


Fig. 14. Visual examples of different stages in the network model. (a) Input bitemporal remote sensing images. (b) Multilevel features captured by the Siamese backbone network. (c) Feature maps obtained from NFAM. (d) Feature maps obtained from IAEM. (e) Difference map obtained through elementwise subtraction. (f) Reweighted feature maps after SAM. (g) Change probability maps. (h) Change map. Samples sourced from the LEVIR-CD dataset.

TABLE X
ABLATION STUDY ON LOSS FUNCTION COEFFICIENTS

λ_1	$\lambda_2 = \lambda_3 = \lambda_4$	LEVIR-CD	GZ-CD	SYSU-CD
1	0	90.23	87.45	82.36
1	0.25	90.46	87.68	82.57
1	0.5	90.72	87.93	82.74
1	0.75	90.98	88.05	82.85
1	1	**91.06**	**88.25**	**82.97**

auxiliary loss weights gradually improved the model's performance, highlighting the crucial role of deep supervision in enhancing the model's generalization ability and overall learning effectiveness.

F. Network Visualization

To better comprehend our network architecture and visually showcase the features at various stages of our network, we conducted visualization of our method's key stages using a pair of bitemporal remote sensing images from the LEVIR-CD dataset. As illustrated in Fig. 14, starting with a pair of dual-time inputs $T1$ and $T2$, the Siamese backbone captures multilevel features $B1$ and $B2$. Subsequently, the NFAM enhances the representation capability of the bitemporal features, resulting in fused adjacent features depicted in the multilevel feature maps $C1$ and $C2$. Following this, the IAEM facilitates more comprehensive interaction and coupling of the bitemporal features $C1$ and $C2$, generating interactive feature maps $D1$ and $D2$. We then obtain the difference feature map E through elementwise subtraction, and progressively aggregate the differential features using our SAM from coarse to fine, acquiring the reweighted feature map F, showcasing a more precise incremental fusion of features after SAM reweighting. Finally, we concatenate and fuse the multiscale features at the decoding stage, yielding the ultimate output feature map Output.

V. CONCLUSION

In this article, we propose an effective change detection network. Tailored to the characteristics of remote sensing image change detection tasks, the algorithm incorporates three modules to enhance accuracy and robustness. Additionally, deep supervision signals are introduced to aid in network training. The approach facilitates comprehensive coupling of bitemporal features and effectively addresses issues related to ambiguous change area boundary identification and small target omission. Extensive experiments on three public datasets demonstrate that our proposed ISDANet achieves outstanding performance and efficiency compared to other SOTA methods. In future research, we plan to explore multisource data fusion methods, such as integrating optical and SAR data, to enhance the precision and robustness of change detection through cross-modal learning.

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